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Assignment No: 2

Title of Assignment: For the above CPS, identify Goals, Requirements (Functional, Non-functional, User Interface) and perform Analysis.

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Assignment No. 2



Aim: Write goal requirements of the CPS
Using Micro tool create User Story
Mapping.

Theory:

Q. What is the need of requirement engineering?
→ Requirement are the one of the most fundamental part of Software engineering, the main goal of the requirement engineering is to get the clear and exact requirement of the customer and to ensure that software solution is a build according to the requirement. Requirement's are gathered at the beginning of SDLC.

Q. What is the goal and what are the functional non functional ; User interface requirements?
→ Functional requirements defines what are the what the features and functions to be required to develop the software, where Non functional requirements are those which are not just specifies features and so on.
Goal is something that is target For software engineering which can be helpfull For the clients. as to

Part II

1) What is User Story?

- User story map is an effectiveness priority that encourage collaboration between teams and it's commonly used in s/w and product development user story mapping is effective technique for reducing product backlog and planning when should be as it required you to use a need structure to prioritize your teams load.

Q.2] How is it different from functional and non functional requirements?

- The user stories to user on the once what the user person using the product want to be able to do traditional requirement focused on what a product should do remaining different one subtree set important.

Q.3] How is it written?

- User Story should be written one or in two sentence capture who the user is what they want and why?
Requirement: It be very detailed and take longer time to write after go into specific detail.

Q.3] format of User story?

User stories are short simple description of feature told from perspective of person derive the new capabilities usually a user or customer of system typically follow template.

as a <type-of-user> I want > something that <Some person>

D - User stories after written on indore work strictly mates stored in arranged on table of facilities planning and discussion as such they strictly planning & discussion as such they strictly the focus feature to discussing them.

Q.4] Application with Agile Methodology

→ In this User story is short, format language description of what user want

D within a software product, a small of users in a agile setting user stories key tool in incremental development,

Conclusion:

→ Hence we have Successfully implemented User stories.

CPS- Freelancer Trying to Manage, Collaborate, Track Changes in Code

GOALS

To streamline code management, collaboration, and version tracking, ensuring efficient workflow and minimal conflicts among contributors.

FUNCTIONAL REQUIREMENTS

- **Version Control Integration** – Support Git and other version control systems for seamless code tracking.
- **Real-time Collaboration** – Allow multiple developers to work on the same codebase simultaneously with live updates.
- **Branching & Merging** – Enable easy branching and merging of code to prevent conflicts and maintain code integrity.
- **Code Review & Comments** – Provide a feature for inline code reviews and feedback.
- **Task & Issue Tracking** – Integrate task management tools to track bugs, feature requests, and work progress.

NON-FUNCTIONAL REQUIREMENTS

- **Performance** – Ensure quick syncing of changes and smooth real-time collaboration.
- **Security** – Implement role-based access control, authentication, and encryption to protect sensitive code.
- **Scalability** – Support projects ranging from small scripts to large-scale enterprise applications.
- **Reliability** – Maintain a 99.9% uptime to ensure uninterrupted access to repositories.
- **Accessibility** – Provide a user-friendly interface with keyboard shortcuts for faster navigation.

USER-INTERFACE REQUIREMENTS

- **Clean & Organized Dashboard** – Present a structured layout with repositories, tasks, and recent activities.
- **Diff View & History Logs** – Show side-by-side code comparisons and detailed commit histories.
- **Drag-and-Drop File Management** – Enable easy upload and organization of project files.
- **Notifications & Alerts** – Keep users updated on changes, reviews, and commits.
- **Dark & Light Mode** – Offer theme customization for comfortable viewing.
- **Code Snippets & Templates** – Allow saving reusable code snippets for efficiency.

GOALS

To streamline code management, collaboration, and version tracking, ensuring efficient workflow and minimal conflicts among contributors.

CPS - Balancing Academic Responsibilities with Personal Time

GOALS

To effectively manage academic workload while maintaining personal time for relaxation, hobbies, and well-being, ensuring a balanced and productive lifestyle.

FUNCTIONAL REQUIREMENTS

- **Smart Scheduling** – Implement a planner that prioritizes academic deadlines while reserving personal time.
- **Task Prioritization** – Provide a to-do list with urgency levels to manage assignments and projects effectively.
- **Time Tracking & Reminders** – Set up reminders for study sessions, breaks, and personal activities.
- **Progress Monitoring** – Offer insights into completed and pending academic tasks for better planning.
- **Customizable Routine** – Allow students to adjust schedules based on changing academic and personal needs.

NON-FUNCTIONAL REQUIREMENTS

- **Performance** – Ensure quick access and smooth functioning of scheduling and tracking tools.
- **Security** – Protect personal data and academic schedules with secure authentication.
- **Scalability** – Support different academic levels, from school to higher education.
- **Reliability** – Maintain a uptime to ensure consistent access to planning tools.
- **Accessibility** – Ensure compatibility with different devices and screen sizes for convenience.

USER-INTERFACE REQUIREMENTS

- **Simple & Intuitive Dashboard** – Present academic tasks and personal time slots in a clear layout.
- **Drag-and-Drop Scheduling** – Allow easy rescheduling of tasks and events.
- **Visual Progress Indicators** – Use graphs and trackers to monitor study hours and leisure balance.
- **Dark & Light Mode** – Offer theme customization for better usability.
- **Notifications & Alerts** – Send reminders for upcoming deadlines and scheduled personal breaks.
- **Sync Across Devices** – Ensure seamless access to schedules from mobile, tablet, and desktop.

CPS - Junior Software Engineer Trying to Fix Errors in My Code and Make It Work Correctly

GOALS

To efficiently identify, debug, and resolve errors in the code, ensuring functionality, performance, and maintainability.

FUNCTIONAL REQUIREMENTS

- **Error Detection & Highlighting** – Provide real-time syntax and runtime error detection.
- **Debugging Tools** – Support breakpoints, step-by-step execution, and variable inspection.
- **Code Suggestions & Autocomplete** – Offer intelligent code suggestions to minimize errors.
- **Logging & Error Reports** – Generate detailed logs to help diagnose issues quickly.
- **Version Control Integration** – Track code changes and revert to previous working versions when needed.

NON-FUNCTIONAL REQUIREMENTS

- **Performance** – Ensure smooth execution of debugging tools without lag.
- **Security** – Safeguard source code with secure authentication and encrypted storage.
- **Scalability** – Support projects of various sizes, from small scripts to large applications.
- **Reliability** – Maintain a 99.9% uptime for debugging and development tools.
- **Accessibility** – Provide keyboard shortcuts and customizable UI for ease of use.

USER-INTERFACE REQUIREMENTS

- **Clean & Organized Code Editor** – Display syntax highlighting and structured indentation.
- **Error Messages & Fix Suggestions** – Show clear explanations and possible fixes for errors.
- **Collapsible Code Sections** – Allow hiding of specific parts of the code for better focus.
- **Dark & Light Mode** – Enable theme customization for better readability.
- **Search & Replace** – Implement powerful search and replace functionality for quick modifications.
- **Execution Console & Output Viewer** – Provide a clear output display for debugging results.

Result:

User Story I : Balance Student academic responsibilities with personal time .

Allocate the task according to free time

Suggest to add one some good habits

Track his academic time table and routine at the home

allocate the task according to his free time

Track the interests of the user through out his daily routine

MVP

As a student I should be able to manage my academics as well as personal routine, So that its very easy to manage the time.

As a student, I should be able to set reminders for upcoming assignments and deadlines, So that no need to remember the all the stuff.

As a student, I should be able to track my study hours and break time, So that I can effectively use my time

As a student, I should receive suggestions to improve my study habits based on my routine, So that I am able to improve my daily routines

As a student, I should be able to integrate my academic calendar with the app for better planning, So that its quite easy to manage to track the daily routines along with the academics.

Backlog

As a student, I should be able to collaborate with classmates for group projects and discussions, so that , so that I can collaboratively manage.

As a student, I should receive motivational quotes or productivity tips daily so that I can be motivated along with the daily routine

As a student, I should be able to set priority levels for tasks, so that tasks with the higher priority must complete first

As a student, I should be able to sync my tasks with Google Calendar, so that I can integratively use the tasks and the app.

User Story II: Freelancer Trying to manage, collaborate and track changes in code.

Create a system that helps freelance web developers efficiently manage their code, minimize merge conflicts, and learn Git best practices.

Git Workflow Guide

Must detect the Merge Conflicts

MVP 2

As a freelance web developer, I should be able to collaboratively work with the team members so that it can be easy to work on same projects.

As a freelance web developer, I can be able to backup my code without remembering all the git command, So that I can be able to backup my code

Backlog 3

As a student, I should be able to collaborate with classmates for group projects and discussions, so that, so that I can collaboratively manage.

As a student, I should receive motivational quotes or productivity tips daily so that I can be motivated along with the daily routine

As a student, I should be able to sync my tasks with Google Calendar, so that I can integratively use the tasks and the app.

User Story II: Freelancer Trying to manage, collaborate and track changes in code.

Create a system that helps junior software engineers efficiently debug their code, understand errors, and learn debugging best practices.

Error Debugging Guide

Must provide Clear Error Explanation

MVP | 2

As a junior software engineer, I should be able to debug my code efficiently with suggestions on how to fix errors, so that I can improve my problem-solving skill.

As a junior software engineer, I should receive real-time syntax and logical error detection, so that I can fix issues before running the code.

Backlog | 3

As a junior software engineer, I should be able to collaborate with senior developers or mentors for debugging assistance, so that I can learn best practices.

As a junior software engineer, I should receive daily coding challenges and debugging exercises, so that I can improve my skills over time.

As a junior software engineer, I should be able to integrate debugging tools with my IDE, so that I can streamline my workflow.

Pranadali
3/2/2025